SO-101
Motion Lab.

From a robot model to testable questions about motion, reach and communication.

Three linked numerical studies.
Original technical explainers.
Complete figures, not cropped previews.

PYTHON + JAVASCRIPT | URDF | AI-ASSISTED LEARNING

November 2026 · Barcelona
Simulation only. No hardware or collision validation.



The question comes before the image.

01

Can the model be checked?

Forward kinematics, numerical IK, joint limits and independent checks.

A tested Python learning toolkit, built with AI coding assistance.

02

Does interpolation draw a line?

Compare joint-space motion with Cartesian tracking under a fixed tool direction.

Keep failed scenarios visible, rather than showing only successful paths.

03

Is a visible target reachable?

Sample legal poses and distinguish a reachable witness from a provably outside-bound target.

Do not confuse sampled points with an exact workspace boundary.

Five arm joints. One gripper.

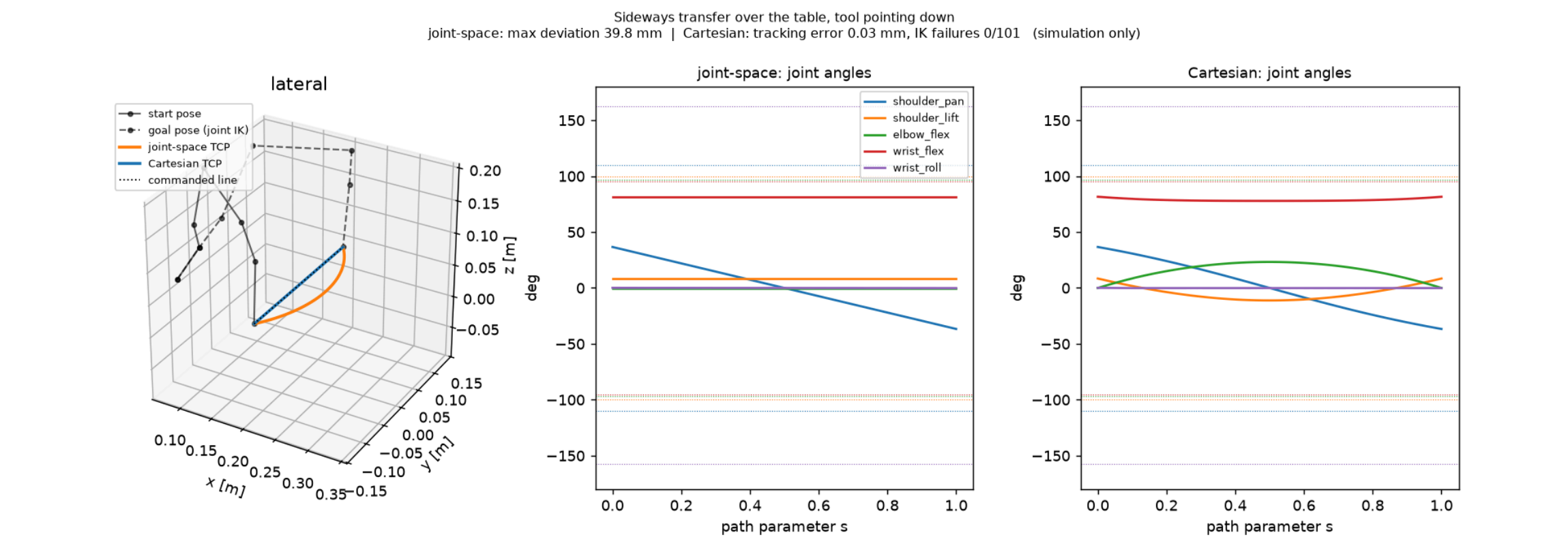


Use the model’s declared frames and limits, not an AI-generated approximation.

- Official **so101_new_calib.urdf**, TheRobotStudio / SO-ARM100.
- Five moving arm joints in the base-to-TCP chain.
- The gripper jaw is separate from this arm kinematics chain.
- TCP: `gripper_frame_link`.
- Forward transforms, a geometric Jacobian and damped least-squares inverse kinematics.

Model revision a758567 · Apache-2.0. Rendered materials are modified. No powered-hardware test.

The same endpoints do not imply the same path.



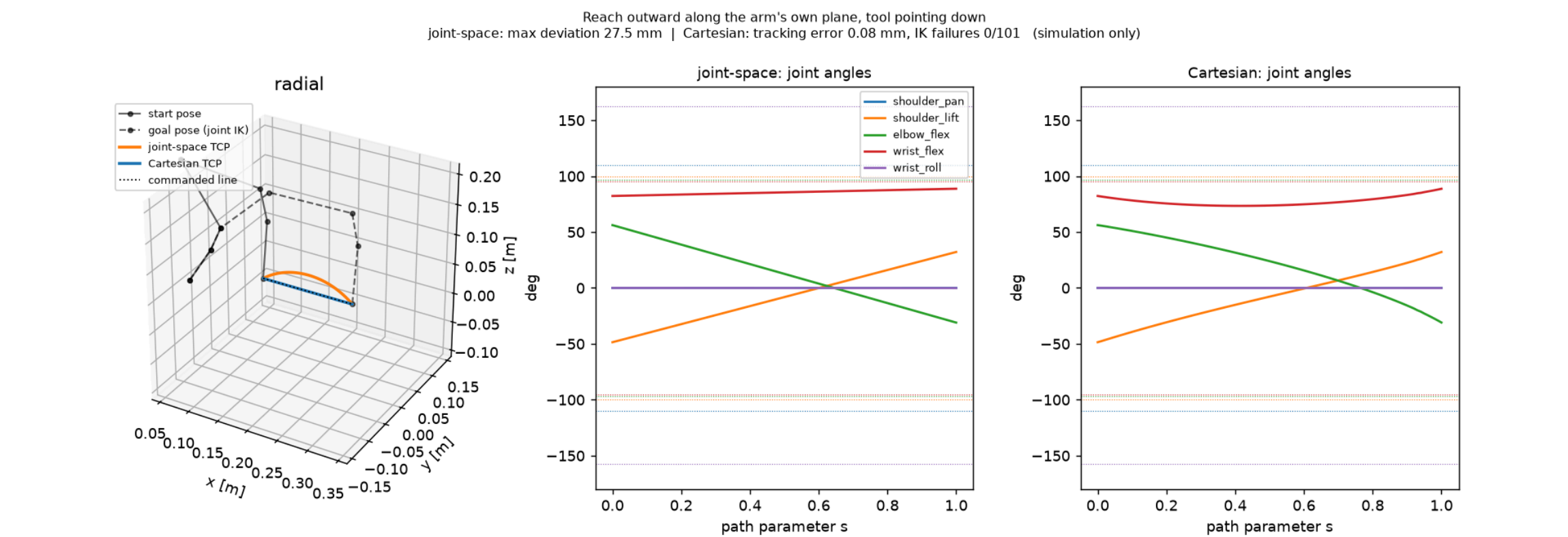
39.8 mm
Joint-space TCP deviation from the line

0.03 mm
Cartesian target tracking error

2.3×
More total joint travel for the Cartesian case

240 mm transfer. The first two values are different metrics, aligned with each method's objective. No collision check.

A second geometry, a different trade-off.



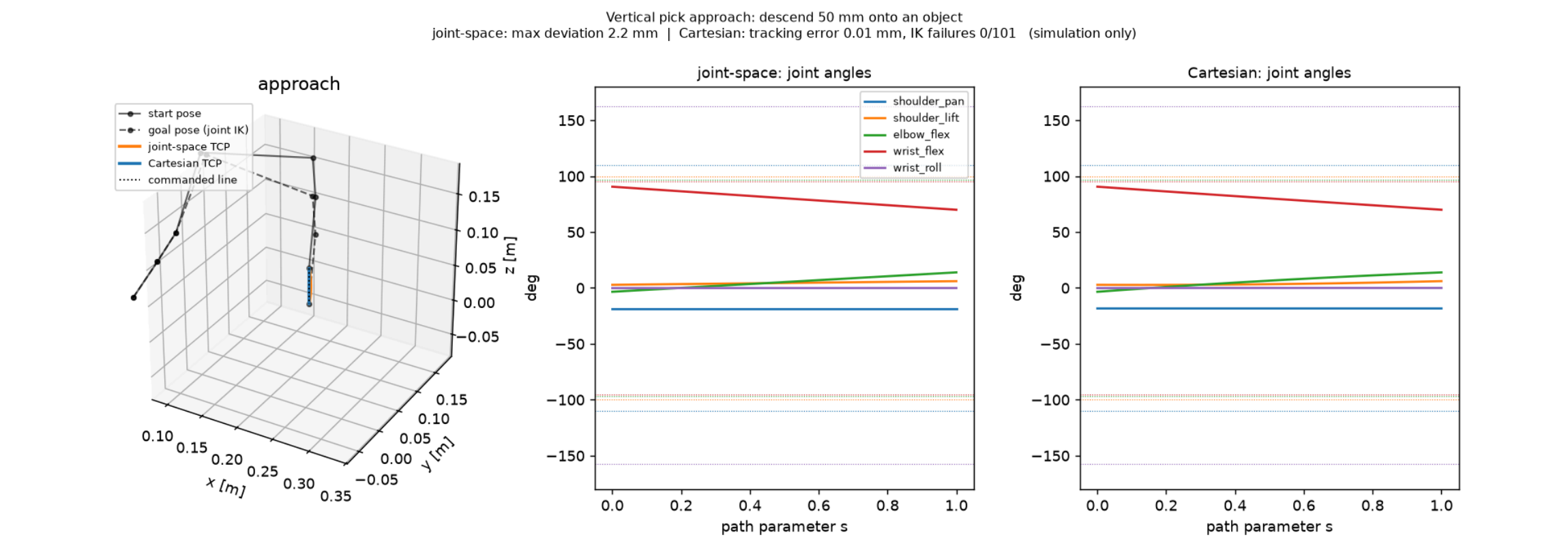
150 mm
Commanded radial transfer

27.5 mm
Joint-space deviation from the line

0 / 101
Failed IK samples in this Cartesian scenario

Cartesian tracking error: 0.08 mm. Joint travel: 3.05 rad versus 3.35 rad. These results are scenario-specific.

Not every move produces a large difference.



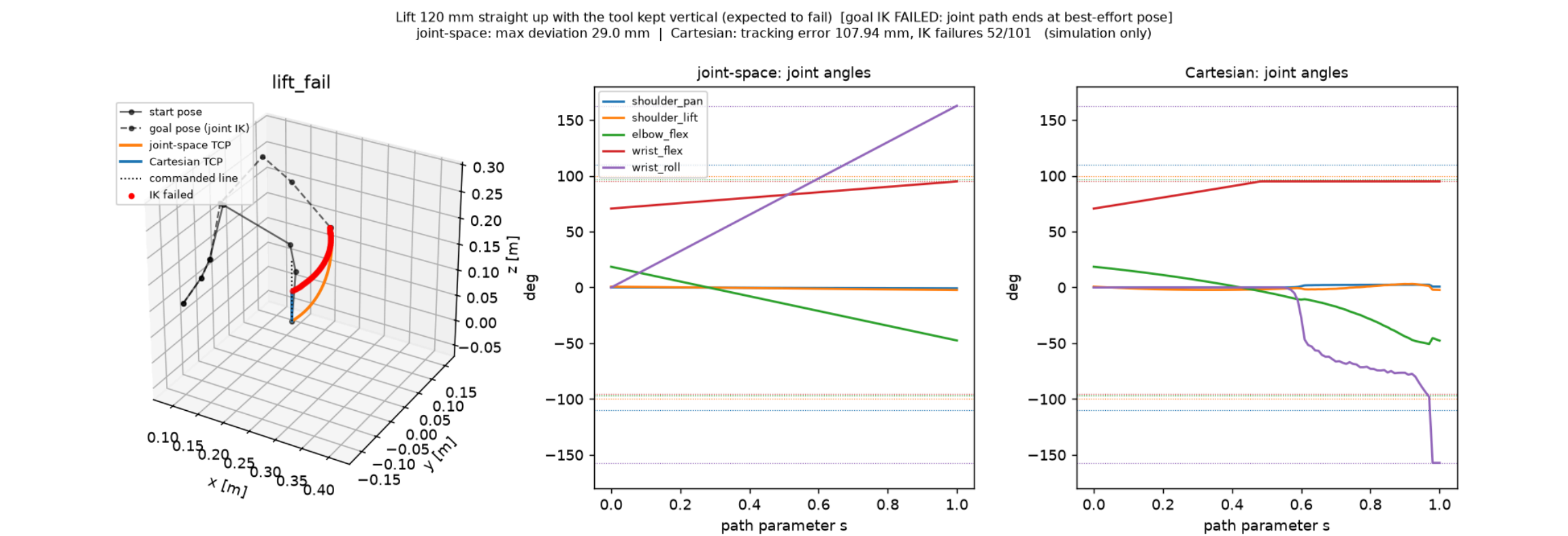
50 mm
Vertical approach

2.2 mm
Joint-space deviation from the line

0.01 mm
Cartesian tracking error

Both methods used about 0.72 rad of total joint travel. A useful study reports when methods behave similarly, too.

A solver failure is part of the result.



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Failed Cartesian IK samples

107.9 mm
Maximum target tracking error

1.03 rad
Largest joint step in the failed sequence

Failure to converge is not, by itself, proof of geometric unreachability. It is not a deployable robot trajectory.

Checks, not just a plausible animation.

Python toolkit · recorded test suite

- **12 passing tests** in the saved study.
- FK compared with a separate quaternion implementation.
- Geometric Jacobian compared with finite differences.
- Joint-limit and unreachable-target handling tested.
- IK benchmark: 200/200 position-only and 192/200 position + direction.

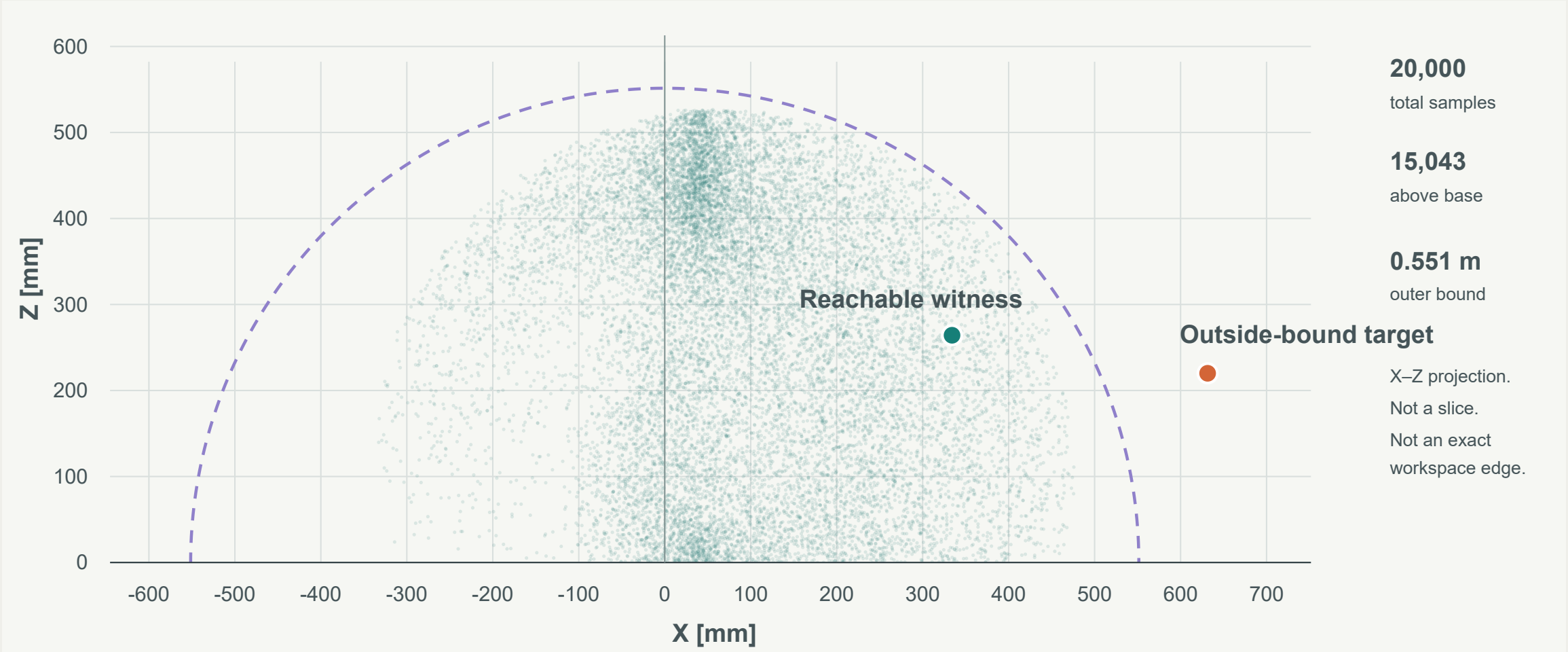
Workspace study · repeated 9 Nov 2026

- **20,000 deterministic samples** reproduced in Aside REPL.
- All sampled joints remained within the URDF limits.
- All TCP samples remained within the conservative outer bound.
- Saved reachable witness reproduced with zero numerical difference.

The standalone CLI rerun was blocked by the current sandbox's path permissions. No new Python-suite pass is claimed today.

> Development was AI-assisted. This is a personal learning and validation project, not a claim of independent professional robotics-software mastery.

A cloud of legal poses is not an exact boundary.



All 20,000 legal configurations were evaluated; 15,043 TCP positions are above the base plane and shown. This is an X–Z projection, not a cross-section.

What can be proved without trusting an IK failure?

CONSERVATIVE BOUND

0.551 m

Sum of the norms of the joint-to-joint translation vectors.

Triangle inequality gives an outer distance bound from the base origin. It is not the arm's exact maximum reach.

CHOSEN OUTSIDE TARGET

0.669 m

Distance of the selected target from the base origin.

About 117 mm beyond the outer bound, so this target is geometrically excluded under the model.

INDEPENDENT AGREEMENT

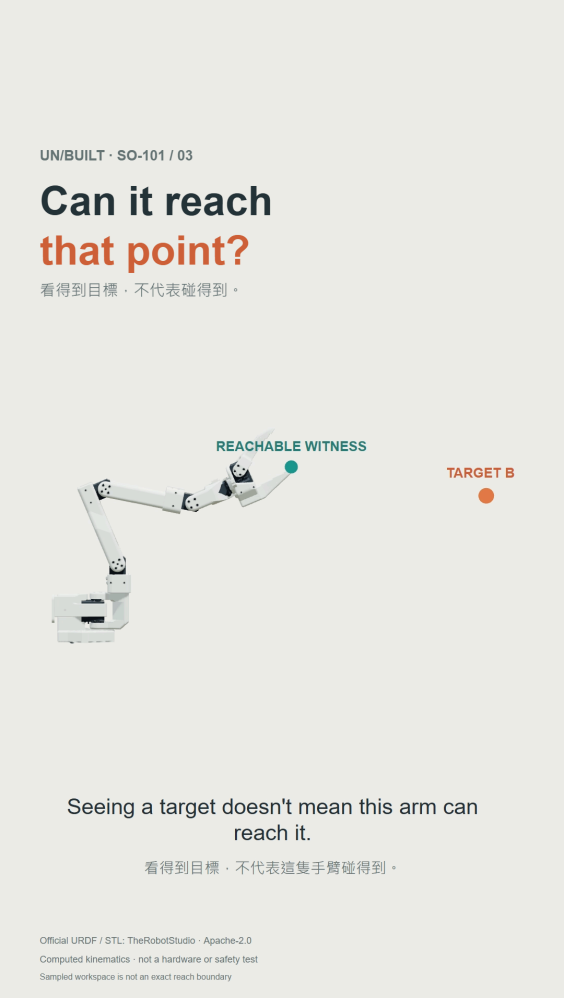
100 poses

Cross-checked against independent Three.js world transforms.

Maximum TCP-position difference: 3.25×10^{-16} m. Numerical agreement is not physical robot accuracy.

A point inside this outer bound is **not automatically reachable**. Only the selected witness is demonstrated by a valid joint configuration.

Make the engineering question legible.



“Seeing a target does not mean the arm can reach it.”

- A 38-second technical Reel connects the model, samples, a reachable witness and an excluded target.
- Original script, synthetic English narration, original procedural music and bilingual captions.
- Geometry and limitations are credited in the published caption.
- Source-based technical visuals, not AI imagery presented as engineering evidence.

Watch the published workspace study

Published 7 November 2026. No hardware, force-control or collision-safety claim.

A precise boundary makes the work stronger.

Completed in this supplement

- URDF-based numerical kinematics and saved test evidence.
- Four scenario comparisons, including a retained failure.
- Reproduced workspace samples, a valid witness and an analytical outer-bound example.
- Original engineering explanation and published technical communication.

Not validated or not yet built

- Powered SO-101 control, camera calibration or measured TCP accuracy.
- Collision-free motion, payload, dynamics or safety.
- A finished independent scan-to-repair perception pipeline.
- A printed and mechanically tested camera mount or end effector.

Next experiment: when hardware is operational, measure the same path with a camera/marker and compare repeatability and TCP error against these predictions.

The evidence travels with the portfolio.

Included evidence

Kinematics toolkit, result figures and reports.
Workspace source, URDF, deterministic samples and QA records.
Full-view figures and this editable-import PDF.

```
Python toolkit:
python -m so101_kin demo
python -m pytest -q

Workspace study:
node reproduce.js
```

Commands are reproduction instructions. Environment/dependency setup is documented in each README; current-run limitations are recorded in QA.

Source and authorship

TheRobotStudio / SO-ARM100
Official SO-101 model · Apache-2.0
Revision a758567c3978…

Chi Lee · personal learning project.
AI-assisted implementation, analysis and production.

Published workspace explanation
LinkedIn · Chi Lee

All figures are shown at their native aspect ratio. No claim of independent software authorship, real robot accuracy or completed future prototypes.